IAM in GCP

Identity and Access Management (IAM) in Google Cloud Platform (GCP)

Overview

Identity and Access Management (IAM) in Google Cloud Platform (GCP) is a framework that enables you to manage **who (identity)** has **what access (roles)** to **which resources**. IAM allows organizations to implement and enforce fine-grained access controls and ensure that the right individuals and services have the appropriate permissions.

IAM is foundational to GCP's security model, offering a unified system for managing access across all GCP services.

Core Concepts

1. **Identities**
    - Represents who is requesting access.
    - Types of identities:
        - Google Account: Individual user accounts.
        - Service Account: Special accounts for applications or virtual machines.
        - Google Groups: Collection of Google accounts.
        - Cloud Identity or G Suite domains: Organizations using Google Workspace.
        - External identities via Identity Federation.
2. **Resources**
    - The target that the identity is trying to access.
    - Examples: Compute Engine VM instances, Cloud Storage buckets, BigQuery datasets.
3. **Roles**
    - Define what permissions an identity has.
    - Types of roles:
        - **Basic roles**: Primitive roles such as Owner, Editor, and Viewer.
        - **Predefined roles**: Granular roles curated by Google (e.g., BigQuery Data Viewer).
        - **Custom roles**: Roles created by users for specific permissions.
4. **Policies**
    - Bindings that connect identities with roles to resources.

- IAM policies are hierarchical and can be applied at the organization, folder, project, or resource level.

Use Cases

1. **Least Privilege Access**
   - Ensures users and services have only the permissions they need, and no more.
2. **Multi-tenancy and Project Isolation**
   - Enables organizations to manage multiple teams and projects securely under a single GCP organization.
3. **Audit and Compliance**
   - IAM policies and logs support regulatory compliance through visibility and control.
4. **Automated Workflows**
   - Service accounts can be used by applications and automation scripts to access resources securely.
5. **Secure API Access**
   - Manage which identities (users or services) can interact with GCP APIs.
6. **Delegated Administration**
   - Teams or departments can manage their resources without needing full administrative privileges.

Benefits of IAM in GCP

1. **Centralized Access Management**
   - Unified system to manage permissions across all GCP services.
2. **Fine-Grained Control**
   - Enables detailed permission management with predefined and custom roles.
3. **Improved Security Posture**
   - Enforces the principle of least privilege, reducing the risk of unauthorized access.
4. **Auditability and Transparency**
   - Integration with Cloud Audit Logs for tracking changes and access attempts.
5. **Scalability**
   - Suitable for organizations of all sizes, from startups to large enterprises with complex hierarchies.
6. **Flexibility**

- Supports cross-project access and hybrid identity scenarios (e.g., on-premises and cloud integration).

7. **Integration with Other GCP Services**
    - IAM is tightly integrated with services like Cloud Storage, BigQuery, Pub/Sub, and more, streamlining access control configuration.

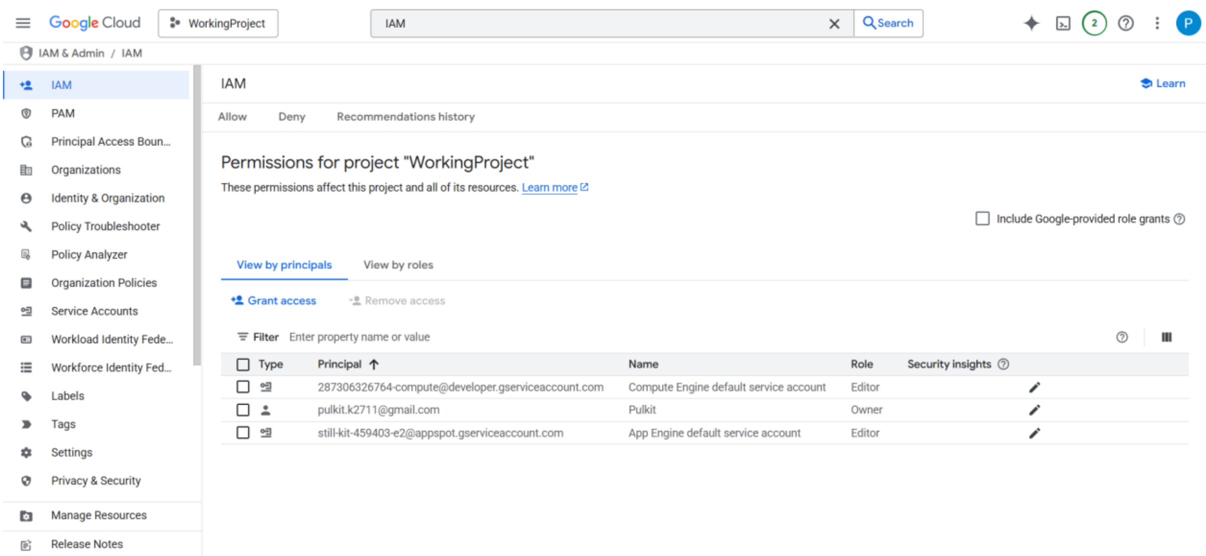
Best Practices

- Use **predefined roles** whenever possible.
- Follow the **principle of least privilege**.
- Regularly **audit IAM policies**.
- Use **service accounts** for applications instead of user credentials.
- Use **organization policies** to set governance rules at scale.
- Enable **Cloud Audit Logs** for visibility into access patterns.

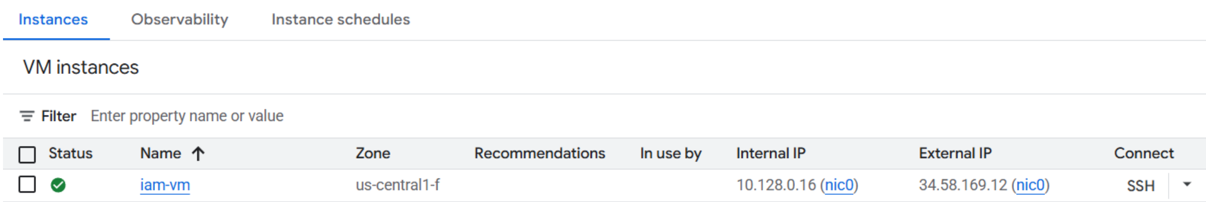
IAM in GCP is a powerful tool that helps organizations manage access securely and efficiently. By using IAM effectively, you can ensure that your cloud resources are protected against unauthorized access while maintaining operational flexibility.

To begin with the Lab

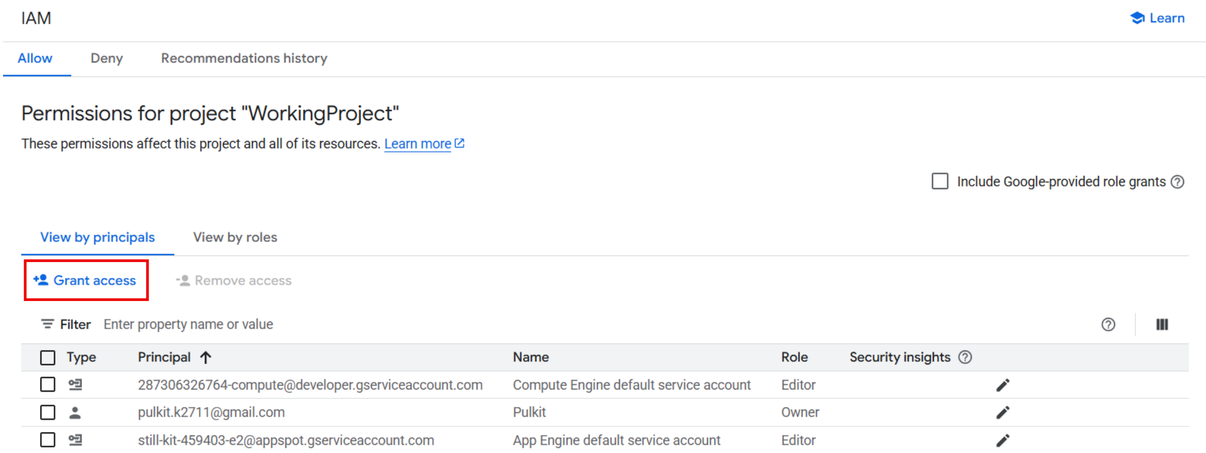
1. From your GCP console, search and navigate to IAM. Here you will see that there are three principal accounts.
2. First is your main account or root account, which you are using to work on GCP, and the others are for Compute Engine and App Engine.



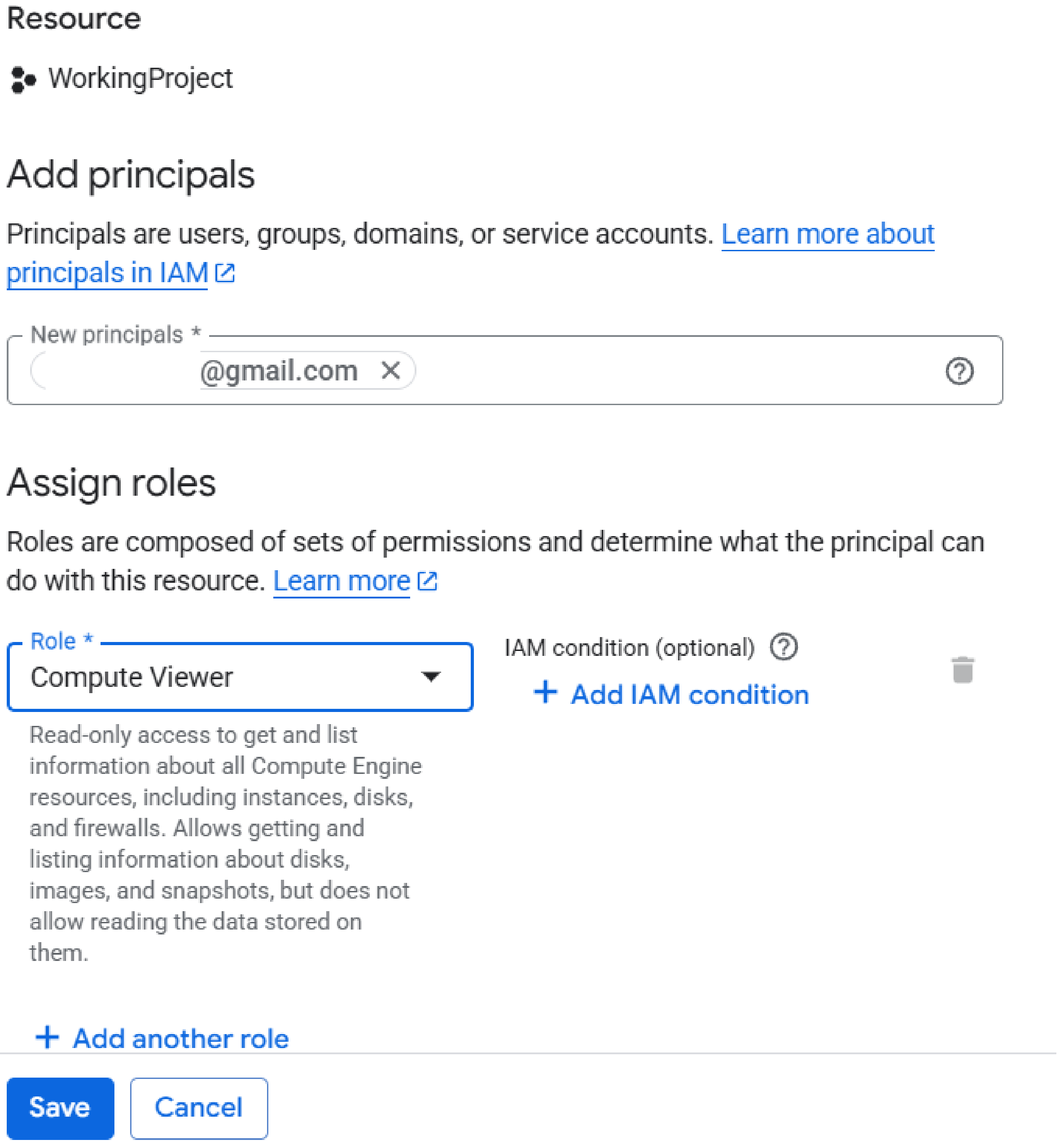
3. Now, to better understand roles and access, we are going to create a Virtual Machine Instance.
4. For that, navigate to EC2 and create a VM. Here you can see that our VM is up and ready.



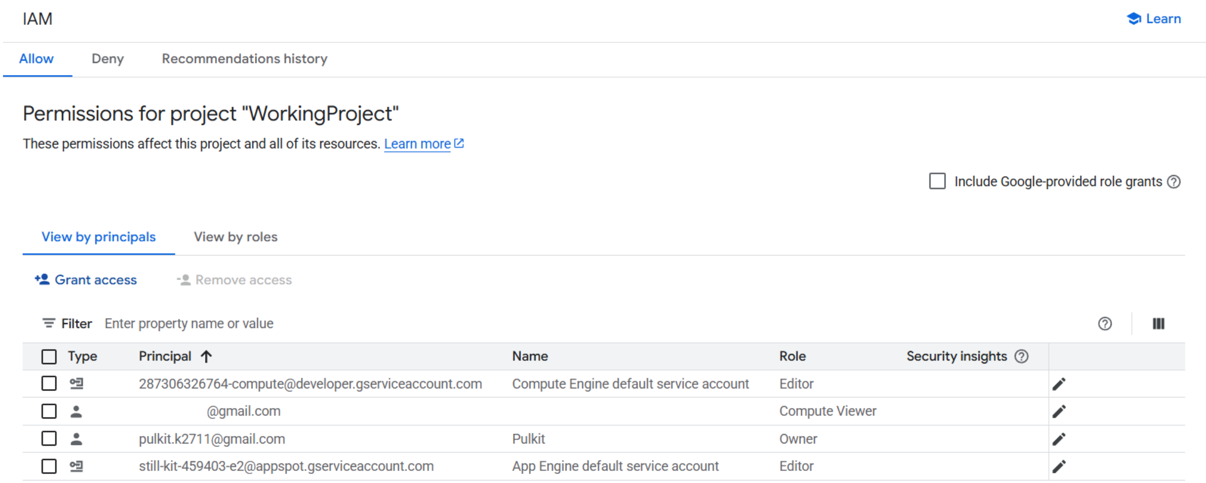
5. Now, come back to IAM and from the IAM dashboard page, click on Grant Access.



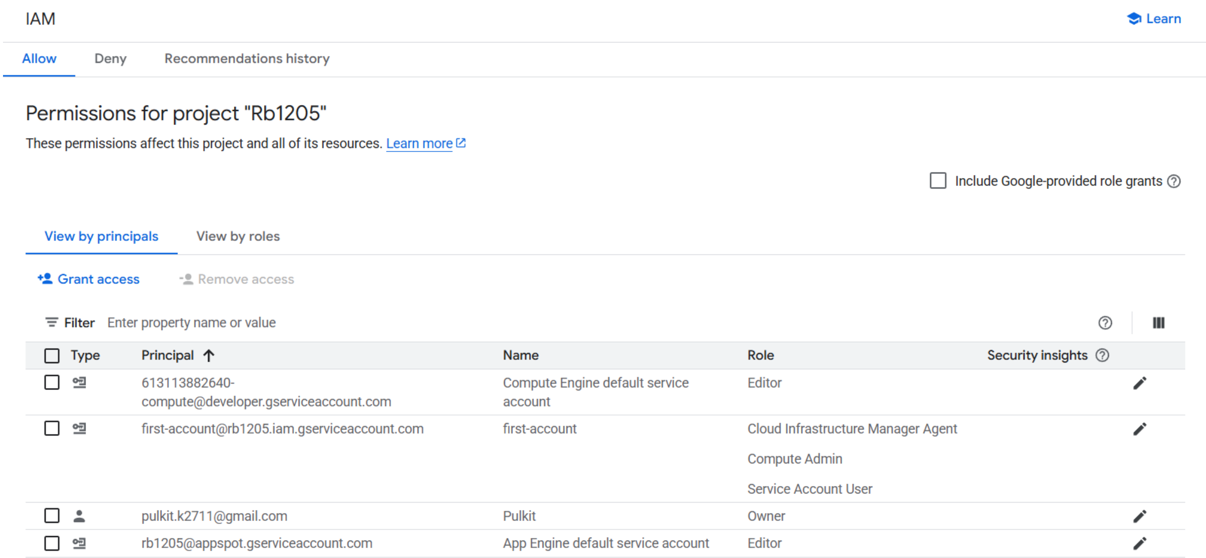
6. In the add principals' area, you need to put an email address to which you want to give access and then you need to assign Compute Viewer role to it. Click on Save.



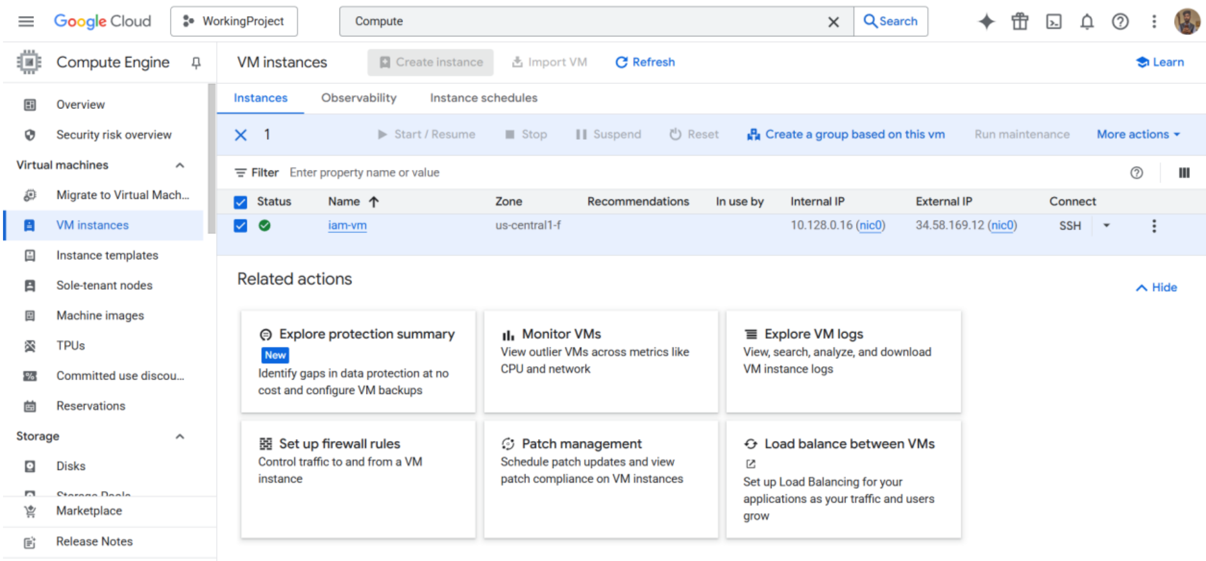
7. In the permission for project area, you can see that our new email address has been added and it has a role attached to it.
8. Also, you can notice that we have given access to the user only to the Working Project.



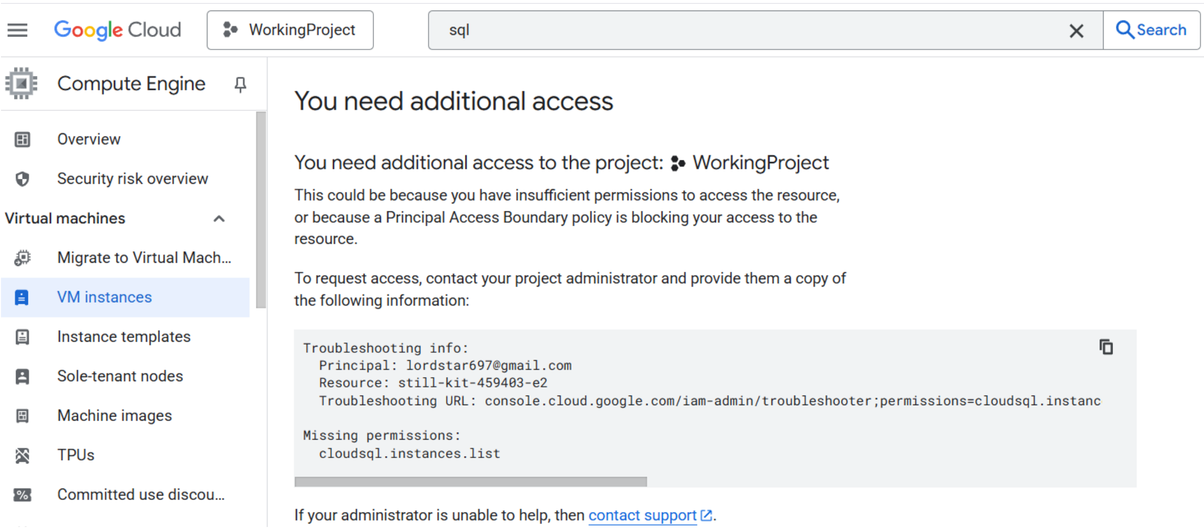
9. If you go to any other project, you will notice that the user and role you added do not exist here.
10. Because the user and role you grant access to exist in another project, not this one.



11. Now, in another browser, we are going to log in to Google Cloud using the same email we have given access to.
12. In another account, we have opened the Google Cloud Console, and we can see that we can view our instances.
13. But if you try to do something with it, then it will deny your access because we have applied only the view role to it.



14. Now, if you try to access any service other than compute, then you will get access denied.
15. So, we tried to open the SQL database service, and we got the message that we'll be needing additional access.



16. Based on the requirements, you can now give access to the user to the services of your choice.
17. Also, you can explore the other options like roles, security, and privacy, etc.